
Detecting and Correcting Reasoning Failures: A Controlled Negative Result for a Residual-Stream Monitor

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Abstract

Language models often solve math problems with fluent reasoning that still ends in a wrong answer. If the model’s internal state signaled the error while the solution was being written, an agent could stop and recompute before committing to it. We test the most direct version of this idea: track how far the hidden state has drifted from where the solution started and, when it drifts too far, roll back a few tokens and prompt the model to recheck its work. We evaluate the monitor by whether the final answer is correct, in the model’s intended prompt format, and against controls that isolate what it adds, including a no-intervention baseline and intervening at a random point. Across the evaluated configurations on two open 7B models and two arithmetic benchmarks, the monitor provides no consistent accuracy improvement. In the two configurations with the most incorrect answers, detection AUC is not distinguishable from chance; false-trigger rates reach 98%, and monitor-guided intervention shows no recovery advantage over random-position intervention. We also show why letting the model judge its own coherence makes such a method look successful when it is not, and we state the conditions a mid-generation monitor must meet before it can be trusted.

1 Introduction

A language model can write a math solution that reads fluently but reaches an incorrect answer: an early mistake can propagate through the remaining reasoning [Cobbe et al., 2021, Lightman et al., 2023]. Post-generation checks assess a completed response [Kuhn et al., 2023, Azaria and Mitchell, 2023]. Detecting a potential error during generation could instead allow an intervention before the solution is complete. One possible source of such a signal is the residual stream, the representation each layer reads and updates. Prior work has identified linearly accessible truth-related information in some model representations [Marks and Tegmark, 2023, Azaria and Mitchell, 2023]. We test whether drift in the residual stream during generation predicts an incorrect final answer.

We test its simplest version: record the hidden state at the start of each solution, measure how far it moves, and when the distance crosses a threshold, roll back 15 tokens, insert “Wait, let me recalculate,” and let the model continue. **(RQ1)** Does the drift score predict a wrong final answer? **(RQ2)** Does acting on it improve final accuracy, compared with the no-intervention baseline and with acting at a random point? We score by the final answer against the reference, not by the model’s opinion of its own work; we prompt each model in its intended chat format, which alone raises Qwen2.5-7B’s GSM8K accuracy from 0.40 to 0.90 and removes the chat-template collapse that raw prompting causes; and we compare every result against the no-intervention baseline, each part of the intervention alone, and a trigger placed at random.

The monitor does not help. Our contributions:

1. **A controlled evaluation of residual-stream drift monitoring for arithmetic reasoning** on two open 7B models, two benchmarks, and two distance measures, with three controls.
2. **Evidence that neither the signal (RQ1) nor the intervention (RQ2) helps.** Accuracy is unchanged or lower than the no-intervention baseline in five of six settings and within noise in the sixth; the trigger recovers no more errors than a random one; in configurations with more observed errors, the detector ranks them no better than chance and fires on up to 98% of correct solutions.
3. **Evidence that a model’s judgment of its own coherence cannot replace answer checking:** high coherence ratings on unchanged incorrect answers show that this judge does not reliably measure error correction. We state the bar a mid-generation monitor must clear.

2 Related work

Mathematical reasoning in language models is scored by final-answer accuracy on benchmarks such as GSM8K [Cobbe et al., 2021] and SVAMP [Patel et al., 2021], and improved by chain-of-thought prompting [Wei et al., 2022], self-consistency [Wang et al., 2023], and step-level verifiers [Lightman et al., 2023]. Self-feedback methods [Madaan et al., 2023, Shinn et al., 2023] let the model grade its own output, which is unreliable without an external signal [Huang et al., 2024] and inherits judge biases [Zheng et al., 2023]. Mechanistic work localizes computation in the residual stream [Geva et al., 2021, Meng et al., 2022, Belrose et al., 2023, Bricken et al., 2023], finds linear structure for truthfulness [Zou et al., 2023, Marks and Tegmark, 2023, Azaria and Mitchell, 2023], and views generation as a trajectory [Elhage et al., 2021, Wang et al., 2022], failure has been framed as gradual drift [Spataru et al., 2024, Halawi et al., 2024]. Inference-time interventions edit decoding or the cache [Chuang et al., 2024, Zhang et al., 2023, Li et al., 2024]. Our evaluation focuses on whether residual-stream drift predicts incorrect arithmetic answers and whether interventions guided by that score improve on sampled baselines, greedy continuation controls, and random-position intervention.

3 Setup

We use two open instruction-tuned 7B models, Qwen2.5-7B-Instruct [Yang et al., 2024] (primary) and Mistral-7B-Instruct-v0.1 [Jiang et al., 2023], on two grade-school word-problem benchmarks, GSM8K [Cobbe et al., 2021] and SVAMP [Patel et al., 2021] (SVAMP on the primary model only). With two distance measures this gives six settings, each with its own random-control run. Detection generations are sampled (temperature 0.85, top- k 50, up to 400 tokens; bfloat16 via `transformer_lens`, seed 42); recovery continuations are greedy, up to 256 tokens; $N=100$ problems per setting. An answer counts as correct only if the extracted final number matches the reference. All primary results use reference-answer correctness. A third model, Yi-6B-Chat [01.AI, 2024], is excluded for the reason given in Section 6.

4 Methodology

Anchor and drift score. At layer ℓ , each generated token t has a residual-stream vector $h_t^{(\ell)} \in \mathbb{R}^d$; we use its direction $\hat{h}_t = h_t^{(\ell)} / (\|h_t^{(\ell)}\|_2 + \epsilon)$. The first $w_H=10$ generated states form a fixed reference anchor for measuring subsequent drift. Each later state receives a drift score. The L_2 score is the squared distance from the current direction to the anchor mean $\bar{h}$, $D_t = \|\hat{h}_t - \bar{h}\|_2^2$. The *Sinkhorn* score compares sets rather than points: the entropy-regularized 2-Wasserstein distance between the w_H anchor states and the $k=5$ most recent states [Cuturi, 2013, Feydy et al., 2019]. Each score is converted to a z -score against the 5 states after the anchor (the tare window), and the monitor fires the first time z exceeds a threshold τ . (A Sinkhorn distance between two single vectors is just the squared distance and only the set-to-set form performs any transport.)

What happens on a trigger. Generation resumes greedily along four paths on the same problems: (A) no intervention; (B) roll back $r=15$ tokens and insert the reset prompt (the full method); (C) insert the prompt without rollback; (D) roll back without the prompt. Clearing the key-value cache changes nothing under greedy decoding, so rollback and prompt insertion are the intervention components.

Random-position control. A separate run applies the full intervention at a uniformly random position after the tare window, without looking at the hidden state. It uses the same seed, so its sampled solutions and wrong-answer set are identical to the monitor run’s; it fires on every problem, so it controls for where the intervention lands, not how often.

Choosing the operating point, and what we measure. The layer (from $\{18, 20, 22, 24, 26\}$) and τ are chosen on held-out data by how well the score separates math from two non-math tasks (Appendix B). For RQ1 we report the AUC of the per-problem maximum z -score for predicting a wrong final answer (0.5 is a coin flip), with n_{wrong} (the wrong answers it is estimated from), n_{genuine} (the triggered wrong answers on which recovery is measured), and the false-trigger rate (the share of correct solutions the monitor fires on). For RQ2 we report final accuracy over all problems and each arm’s lift on the genuine-error subset: its wrong→right flips minus its own do-nothing path’s.

5 Results

RQ1: the drift score does not detectably predict a wrong answer. Calibration found no good operating point: one setting met the target only at its boundary and failed validation (Qwen GSM8K Sinkhorn), the rest fell back to the best available separation, and Mistral GSM8K L2 ended at the most permissive threshold ($\tau = 2$, fires on 99/100), hence its 0.98 false-trigger rate (Appendix B). Table 1 reports each setting with the number of wrong answers it contains. Only the two Mistral settings (56 wrong of 100) give a usable estimate, and both are indistinguishable from a coin flip (0.514 and 0.531; permutation test over 10,000 label shuffles, $p = 0.82$ and 0.60). The 20 problems used for layer selection are among the 100 evaluated; on the 80 disjoint problems the AUCs are 0.517 (L2, bootstrap 95% CI $[0.40, 0.65]$) and 0.500 (Sinkhorn, $[0.37, 0.62]$), so the overlap does not change the picture (Appendix B). The L2 monitor fires on 48–98% of *correct* solutions. The two distance measures agree (AUC 0.53 vs. 0.51; accuracy within 0.07), so comparing sets instead of points does not help.

Table 1: Detection and final accuracy ($N = 100$ per setting). FT: false-trigger rate. $n_{\text{wrong}}/n_{\text{gen}}$: triggered baseline errors. S: sampled baseline. C: prompt only. B: full intervention. Random: random position intervention.

Model	Dataset	Metric	Detection				Final accuracy			
			AUC	n_{wrong}	n_{gen}	FT	S	C	B	Random
Qwen2.5-7B	GSM8K	Sinkhorn	0.62	10	2	0.17	0.90	0.89	0.89	0.92
Qwen2.5-7B	GSM8K	L2	0.46	10	5	0.48	0.90	0.87	0.87	0.92
Qwen2.5-7B	SVAMP	Sinkhorn	0.67	7	6	0.89	0.93	0.91	0.93	0.91
Qwen2.5-7B	SVAMP	L2	0.55	7	6	0.96	0.93	0.93	0.91	0.91
Mistral-7B	GSM8K	Sinkhorn	0.51	56	35	0.68	0.44	0.46	0.46	0.37
Mistral-7B	GSM8K	L2	0.53	56	56	0.98	0.44	0.44	0.39	0.37

The failures are real arithmetic errors, not formatting collapse. Without the chat template, most failures are template collapse (the model emits formatting tokens and stops reasoning); with it, classifying every triggered wrong answer by its generated text, no setting contains one. On Mistral, 50/56 (L2) and 29/35 (Sinkhorn) failures are complete solutions with a wrong answer, the rest never reach an answer; on Qwen, 15/19. This classification reads the greedy continuation from the trigger point (the sampled text was not stored), and 6 of the 50 and 5 of the 29 continuations are in fact correct; it characterizes the failure mode the monitor acts on, not the original sampled errors themselves (Appendix D). The signal is not empty: on Mistral Sinkhorn it separates math from summarization by $z_{\text{gap}} = 5.10$ while ranking correct against incorrect math at 0.514; it tracks what task the model is doing, not whether it is doing it right.

RQ2: acting on the score does not detectably beat doing nothing or a random trigger. Table 1 reports final accuracy over all 100 problems in each setting, which is the number an agent would actually see. Two reference points are involved and must be kept apart. The *sampled baseline* is the accuracy of the original sampled solution with no intervention at all (0.44 on Mistral GSM8K, 0.90–0.93 on Qwen); this is the Table 1 column labeled S. The *greedy continuation control*, path A in

Section 4, restarts greedy decoding from the trigger point with no rollback and no prompt; because it is a fresh decode, it fixes some errors on its own, and on Mistral it reaches 0.41 (L2), 0.45 (Sinkhorn) and 0.40 (random run) over all 100 problems. Against the sampled baseline, the full method is unchanged or lower in five of six settings and higher by 0.02 in the sixth, and every Mistral Wilson 95% interval contains the 0.44 baseline (Appendix F); on both Qwen GSM8K settings the random-position trigger is nominally the best path (0.92 vs. 0.90), a difference of two problems. Against its own continuation control, the paired full-minus-control difference, bootstrapped over problems, is -0.02 $[-0.11, +0.07]$ on L2, $+0.01$ $[-0.06, +0.08]$ on Sinkhorn and -0.03 $[-0.11, +0.05]$ on the random run; each interval straddles zero. The intervention also has a cost the headline hides: the full method turns 14, 5 and 13 initially correct answers wrong in those three runs, comparable to the number it fixes (9, 7 and 6).

The recovery question is narrower: among problems the sampled baseline got wrong, how many does each path turn right, and does the monitor’s choice of *where* to intervene add anything over a random position? We report lift, each arm’s wrong→right flips minus its own continuation control’s, because the control already repairs some errors. On Mistral L2 the full method flips 9 of 56 wrong answers against the control’s 6 (lift +3), and the random trigger flips 6 against 3 (also +3). On Sinkhorn the monitor fires on only 35 of the 56 errors, so its +2 against random’s +3 over all 56 compares two policies that intervene on different problems; restricted to the 35 problems both intervene on, the monitor’s lift is +2 and random’s is -1 , a matched-timing comparison on too few cases to separate them (Appendix F). Trigger positions are consistent with this: on Mistral L2 the median trigger is token 17, immediately after the 10-token anchor and 5-token tare window and before any arithmetic has appeared, so the “recovery” is a restart of the solution rather than a correction of a located error; in the other settings the median trigger falls between tokens 25 and 69 (Appendix G). Equal lift does not prove equivalence, but using the hidden state recovers no detectably more errors than a random point, in line with evidence that models do not correct their own reasoning without outside feedback [Huang et al., 2024]. Sweeping τ on Qwen GSM8K L2, accuracy returns to the 0.90 baseline once the threshold is high enough that the monitor fires on at most 45% of problems (Appendix C).

A model judging its own coherence cannot score recovery. Recovery is often scored by asking the model whether its output stayed coherent. We scored our paths that way, a “sound”/“drifted” verdict from the model under test without the reference (Appendix E, Figure 3). On Mistral it reports 82–95% “recovery” on paths whose real wrong→right rate is 0.05–0.20, and across all six settings it scores *doing nothing* at 0.76–0.91, including 0.91 on Mistral GSM8K L2 where accuracy is 0.44. The score does not track accuracy across models (0.91 for a 0.44-accurate model, 0.85 for a 0.90-accurate one) and rates Qwen’s intervened paths below its baseline (0.55–0.72 vs. 0.76–0.85). The judge rate is computed over the cases with a parseable verdict (55 of 99 triggered cases for path A on Mistral L2) and the correction rate over the 56 originally wrong cases; the run did not store per-case verdicts, so the two cannot be joined on a common set, and the comparison is between population rates, not paired outcomes. A judge that calls doing nothing a success cannot substitute for checking the answer.

6 Limitations

Our evidence covers short arithmetic reasoning on instruction-tuned models without a long-reasoning mode. Qwen2.5-7B provides four settings, but at 0.90–0.93 accuracy it yields only seven to ten wrong answers per setting, resulting in wide AUC confidence intervals; the configurations with the most observed errors are therefore both Mistral-7B on GSM8K. A test that cannot reject chance does not prove chance, so our claim is *no detectable signal at this sample size*. Mistral’s low baseline is genuine (50/56 failures are complete solutions with a wrong answer); Yi-6B-Chat (baseline 0.04) is excluded because 64/80 of its failures never state an answer. The 256-token recovery budget roughly doubles truncation under the intervention on Mistral (9/99 to 18/99). This difference may disadvantage the intervention and limits conclusions about its effectiveness with longer recovery budgets. The layer and threshold were chosen by how well the score separates math from non-math tasks, not correct from incorrect solutions; the threshold-free AUC does not depend on τ , but the layer does, and correctness-based layer selection and a supervised probe on the same activations were not run. Both are the first things to try, together with a task on which the stronger model still fails.

These results do not rule out internal signals of arithmetic failure. They show that the drift scores and interventions evaluated here provide no reliable detection or correction benefit under the tested conditions, and they set the bar such a monitor must clear: (i) score by the final answer against the reference, in the model’s own prompt format; (ii) beat doing nothing on the full task; (iii) beat a random-position trigger on the genuine-error subset, as lift over each arm’s own continuation control; (iv) rank failures above chance where there are enough failures to estimate it; and (v) still work after removing each component. Since capable small models are close to ceiling on GSM8K, this kind of evaluation should use a task they still fail. The monitor studied here clears none.

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A Code availability

The code, run artifacts, and figure scripts are available at <https://github.com/ArnavKakani/DCRF>.

B Layer and threshold selection

The layer is chosen by $z_{\text{gap}} = \overline{\text{max}} z_{\text{target}} - \overline{\text{max}} z_{\text{control}}$ over 20 target and 2×20 control problems (CNN/DailyMail, MBPP), one rollout, and the threshold on a parity split of those traces. Selected operating points are in Table 2. The 20 layer-selection problems are problems 0–19 of the evaluated 100; detection AUC on the 80 remaining problems is 0.517 for L2 (43 wrong; bootstrap 95% CI [0.40, 0.65]) and 0.500 for Sinkhorn ([0.37, 0.62]).

Table 2: Selected operating points. TPR/FPR are trigger rates on reasoning vs. control tasks at the selected τ (task separation, not error separation). Five of six cells miss the target and fall back to best separation; the sixth meets it only at the boundary. Bootstrap 95% AUC intervals for the Qwen cells span 0.27–0.93.

Cell	Layer (z_{gap})	τ	fit TPR	fit FPR	met target
Qwen GSM8K L2	18 (0.016)	6.0	0.70	0.60	no
Qwen GSM8K Sinkhorn	22	15.0	0.60	0.20	borderline
Qwen SVAMP L2	26	4.0	1.00	0.80	no
Qwen SVAMP Sinkhorn	26	6.0	1.00	0.65	no
Mistral GSM8K L2	18 (0.578)	2.0	1.00	0.90	no
Mistral GSM8K Sinkhorn	20 (5.10)	6.0	1.00	0.45	no

C Threshold sweep and seed robustness

Qwen GSM8K L2, 20 problems per threshold: final accuracy 0.75 ($\tau=2$), 0.70 (4), 0.75 (6), 0.90 (8, 10, 12.5, 15); trigger rate falls from 1.00 to 0.10 over the same grid (Figure 1). Trigger rate over five seeds (30 problems each) is 0.57 ± 0.04 .

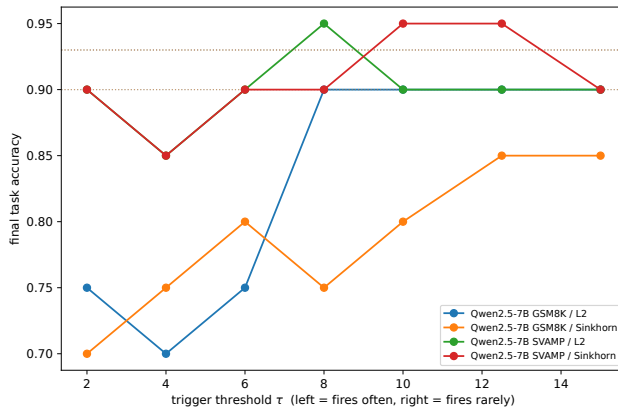


Figure 1: Final accuracy vs. trigger threshold on Qwen GSM8K L2 ($n=20$ per point). Accuracy returns to the 0.90 baseline only once the detector stops firing ($\tau \geq 8$).

D Failure typology

Triggered baseline failures classified on the generated text with the prompt stripped and the model’s end-of-turn token removed. “Wrong answer”: a well-formed solution stating a wrong final answer. “No answer”: looping or cut off by the budget. The text classified is path A, the greedy continuation

from the trigger point; the original sampled generation was not stored. Of the Mistral “wrong answer” continuations, 6/50 (L2) and 5/29 (Sinkhorn) are correct under the extractor. Table 3 and Figure 2 give the counts.

Table 3: Typology of triggered baseline failures, classified on the generated text with the prompt stripped and the end-of-turn token removed. No cell contains a template collapse; Mistral’s failures are overwhelmingly wrong answers, Yi’s are non-completion.

Cell	<i>n</i>	Wrong answer	No answer	Template collapse
Qwen GSM8K L2	5	3	2	0
Qwen GSM8K Sinkhorn	2	1	1	0
Qwen SVAMP L2	6	6	0	0
Qwen SVAMP Sinkhorn	6	5	1	0
Mistral GSM8K L2	56	50	6	0
Mistral GSM8K Sinkhorn	35	29	6	0
Yi GSM8K L2 (excluded)	80	16	64	0

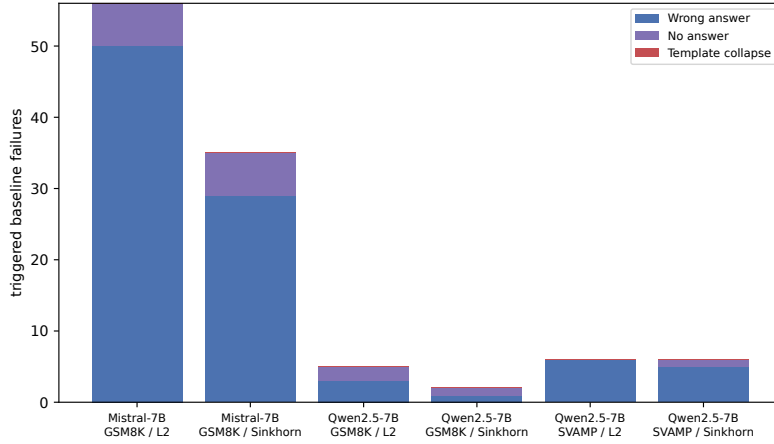


Figure 2: Typology of triggered baseline failures, as in Table 3: wrong answers dominate on Qwen and Mistral, non-completion on Yi, template collapse nowhere.

E Coherence judge vs. answer correctness

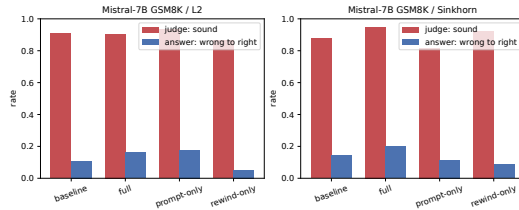


Figure 3: A model judging its own coherence vs. the true answer on the same generations (Mistral). The judge awards 0.82–0.95 “recovery” to every path, including doing nothing, while the real wrong→right rate is 0.05–0.20. Judge rates are over cases with a parseable verdict (Mistral L2: A 55, B 52, C 44, D 59 of 99 triggered); correction rates are over the 56 originally wrong cases. Per-case verdicts were not stored, so the two populations cannot be joined.

F Matched recovery comparison

The random-control run reproduces the detector run’s sampled trajectories (identical seed), so the baseline-wrong sets coincide exactly (Mistral: the same 56 problems). Wrong→right flips on that set are in Table 4:

Table 4: Wrong→right flips per path on the shared baseline-wrong set. Wilson 95% intervals on Mistral final accuracy: full L2 [0.30, 0.49], full Sinkhorn [0.37, 0.56], random [0.28, 0.47], all containing the 0.44 baseline. A is the greedy continuation control, not the sampled baseline. Lift $B-A$: detector +3 vs. random +3 on Mistral L2; +2 vs. +3 on Sinkhorn over all 56 errors (a policy comparison, since Sinkhorn triggers on 35 of them); on the 35 shared cases, monitor A 5, B 7 (+2) vs. random A 3, B 2 (−1). Paired full-minus-control accuracy differences over all 100: −0.02 [−0.11, +0.07] (L2), +0.01 [−0.06, +0.08] (Sinkhorn), −0.03 [−0.11, +0.05] (random). Truncation at the 256-token recovery budget on Mistral L2: A 9/99, B 18/99, C 16/99, D 16/99; random run A 4/100, B 8/100.

Cell	A	B	C	D	Random	
	Control	Full	Prompt	Rollback	A	B
Mistral GSM8K L2 ($n=56$)	6	9	10	3	3	6
Mistral GSM8K Sinkhorn ($n=35$)	5	7	4	3	(3/56)	(6/56)
Qwen GSM8K L2 ($n=5$)	1	1	1	1	(3/10)	(3/10)

G Trigger positions

Figure 4 shows where the monitor fires in each setting. The anchor (10 tokens) and tare window (5 tokens) end at token 15. Median trigger index: Mistral GSM8K L2 17 (65 of 99 triggers at or before token 20; 19 at exactly 15), Mistral GSM8K Sinkhorn 31, Qwen GSM8K L2 67, Qwen GSM8K Sinkhorn 69, Qwen SVAMP L2 25, Qwen SVAMP Sinkhorn 47.

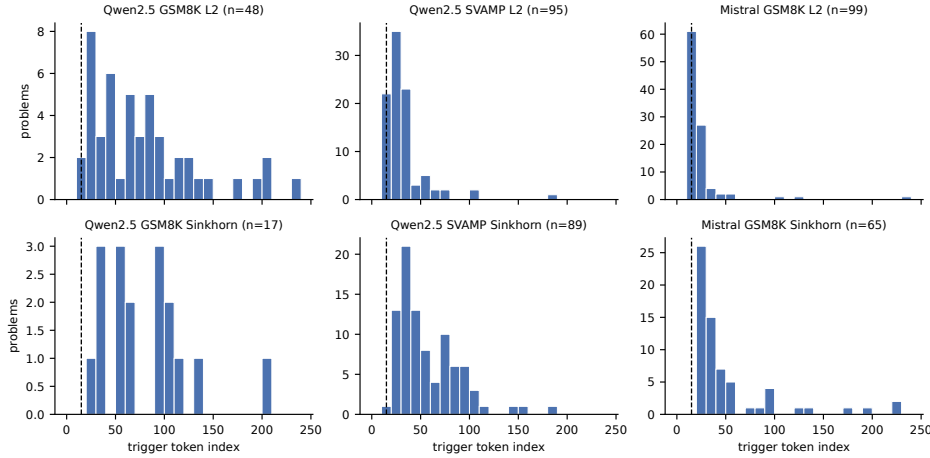


Figure 4: Trigger token index by setting. The dashed line marks index 15, the reported boundary of the anchor and tare windows. On Mistral GSM8K L2, 65 of 99 triggers occur at or before index 20; median trigger indices in the other settings range from 25 to 69. The histogram measures trigger position, not whether an arithmetic operation has occurred.

H Before and after examples

Two genuine-error cases from Mistral-7B GSM8K L2 (prompt stripped; generated text verbatim). In both, the detector fires at token 17–18, immediately after the tare window and before any arithmetic

has been performed, so the “recovery” is a fresh derivation from the start of the solution. That is why a random position does as well.

Recovered (gold 44; spike at token 17). *Q:* Gene is sewing a quilt out of old souvenir t-shirts. He has one shirt from each vacation he has been on. Every shirt is its own quilt block. Each row is made of blocks from a different year of vacations. He goes on four vacations a year and has been vacationing since he was 23 years old. He is now 34. How many quilt blocks does he have in total?

Do nothing (A), wrong: “...in 34 years (the age he is now), he would have gone on a total of $34 \times 4 = 136$ vacations. ... The answer is 136.”

Full method (B), right: “Let’s \n Wait, let me recalculate this step to be sure: 1. From 23 to 34, Gene has been on vacation for 11 years. 2. ... $11 \times 4 = 44$ vacations. ... The answer is 44.”

Not recovered (gold 14; spike at token 18). *Q:* Raymond and Samantha are cousins. Raymond was born 6 years before Samantha. Raymond had a son at the age of 23. If Samantha is now 31, how many years ago was Raymond’s son born?

Do nothing (A), wrong: “... Raymond had a son at the age of 23. So, Raymond’s son was born 23 years ago. The answer is 23 years ago.”

Full method (B), still wrong: “Let’s \n Wait, let me recalculate this step to be sure: 1. Raymond was born 6 years before Samantha. 2. Samantha is now 31. 3. Therefore, Raymond was born $31 - 6 = 25$ years ago. The answer is 25.”

I Ethics statement and reproducibility

This work uses public benchmarks and open models and involves no human subjects. It reports a negative result with its uncertainty stated, and is intended to support careful reporting of inference-time reliability claims for mathematical agents; we foresee no negative impact from a negative result on public benchmarks.

Each run writes a self-contained directory with a manifest (model, dataset, full configuration snapshot, per-stage status) and all intermediate artifacts (trajectories, ablation paths, labels, judge outputs). Every table and figure regenerates from those artifacts without re-running the models; the code and artifacts are at <https://github.com/ArnavKakani/DCRF> (Appendix A). One known limitation: the answer extractor disagrees with a strict “answer is N ” parse on 29 of 3,524 recovery texts, most of them negative dollar amounts written as $-\$N$; none of the correctness outcomes we checked is affected.

All runs executed on single NVIDIA H100 and A100 instances rented from Lambda, one 7B model in `bf16` at a time, over roughly 19 wall-clock hours for the eleven runs (June 24–25, 2026). Assets: GSM8K and SVAMP (MIT license), Qwen2.5-7B-Instruct and Mistral-7B-Instruct-v0.1 (Apache-2.0), Yi-6B-Chat (Yi model license), `transformer_lens` and `geomloss` (MIT).

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